

Course Name: Course Code:	Basic of Electrical Engineering Laboratory (BEEL) 316U06L102	Semester:	I
Date of Performance:	/ /20--	Batch No:	F1
Student Name:	Siddhesh Sarode	Roll No:	
Faculty Sign & Date:		Grade/Marks:	/

Experiment No: 1**Title: Study of types of Electrical components and Instruments.****Aim and Objective of the Experiment:**

1. To understand the working principle of various components and Instruments.
2. To understand the applications of various components and Instruments such as Bread Board, Resistors, Capacitors, Inductors, Cathode Ray Oscilloscope and Function Generator.

COs to be achieved:

CO1: Analyze resistive networks excited by DC sources using various network theorems.

Theory:

A breadboard is a rectangular plastic board with a bunch of tiny holes in it. These holes let you easily insert electronic components to assemble an electronic circuit, like this one with a battery, switch, resistor, and electronic devices like diode, transistor, ICs, etc. The connections are not permanent, so it is easy to remove a component if you make a mistake, or just start over and do a new project. Remember that the inside of the breadboard is made up of sets of five metal clips. This means that each set of five holes forming a half-row (columns A–E or columns F–J) is electrically connected. For example, that means hole A1 is electrically connected to holes B1, C1, D1, and E1. It is not connected to hole A2, because that hole is in a different row, with a separate set of metal clips. It is also not connected to holes F1, G1, H1, I1, or J1.

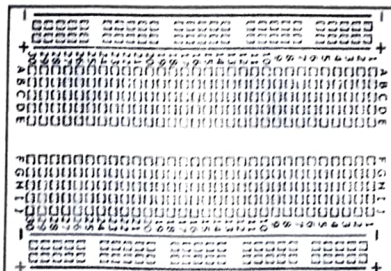


Fig. Bread board

Apply a known amplitude and frequency of sine/triangular/square waveform from signal generator and connect it to one of the channels of the CRO for measurement.

Table 1

Color code of Resistor/Capacitor	Theoretical Value	Practical Value (DMM)
Red-Violet-Brown-silver	$270 \pm 10\%$	255 pF
Red-brown-orange-gold	$21 \times 10^3 \pm 5\%$	21310 pF
Brown-Blue-Violet-silver	$16 \times 10^7 \pm 10\%$	163 μ F

Table 2

Waveform Nature	Theoretical	Measured frequency (Hz)	Theoretical	Measured amplitude peak to peak (V)
Sine wave	1 kHz	1 kHz	4	4
Square wave	1.3 kHz	1.28 kHz	3.80	3.77
Triangular wave	2 kHz	2 kHz	2	2

Post Lab Subjective:

1. State the functions of C.R.O

The functions of C.R.O are as follows -

- CRO is used to display, measure and analyse various types of waveforms and other phenomena in electrical and electronic circuits.
- CROs are used to analyse the waveforms, transient phenomena and other time-varying quantities from a very low-freq range to radio freq.
- CRO provides accurate time and amplitude measurements of voltage signals over a wide range of frequencies.
- CRO is a test instrument which allows an individual to "plot" and "view" two-dimensional graphs of electronic signals.

Conclusion:

By virtue of this experiment, I understood practically, how to correctly decode the value of a register by the bands printed on it. I also got introduced to a CRO and understood the basics of the device and the 3 waveforms it displays.

Signature of faculty in-charge with Date: